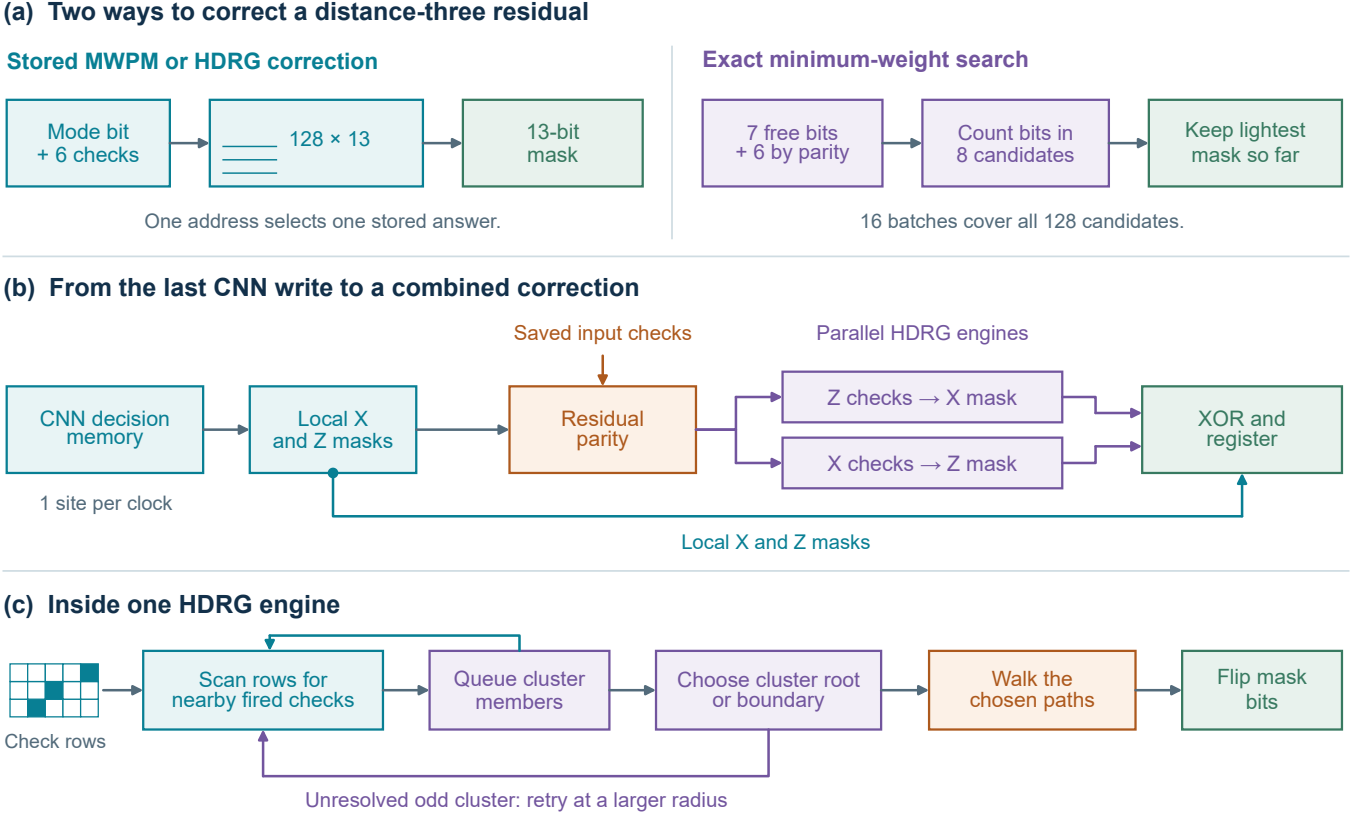

(a) Two ways to correct a distance-three residual
Stored MWPM or HDRG correction
Mode bit + 6 checks
128 × 13
13-bit mask
One address selects one stored answer.
Exact minimum-weight search
7 free bits + 6 by parity
Count bits in 8 candidates
Keep lightest mask so far
16 batches cover all 128 candidates.
(b) From the last CNN write to a combined correction
Saved input checks
Parallel HDRG engines
CNN decision memory
Local X and Z masks
Residual parity
Z checks → X mask
X checks → Z mask
XOR and register
1 site per clock
Local X and Z masks
(c) Inside one HDRG engine
Check rows
Scan rows for nearby fired checks
Queue cluster members
Choose cluster root or boundary
Walk the chosen paths
Flip mask bits
Unresolved odd cluster: retry at a larger radius